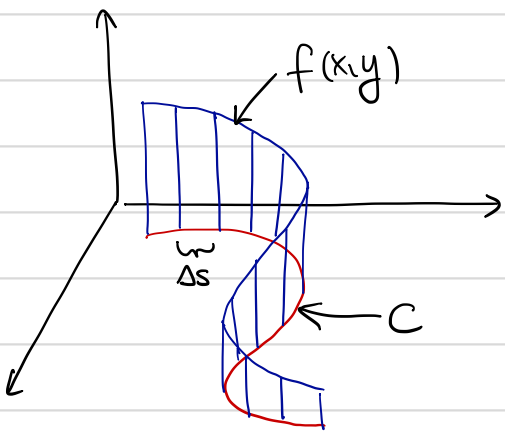
Lecture 13

- Line integrals, Work (16.2)

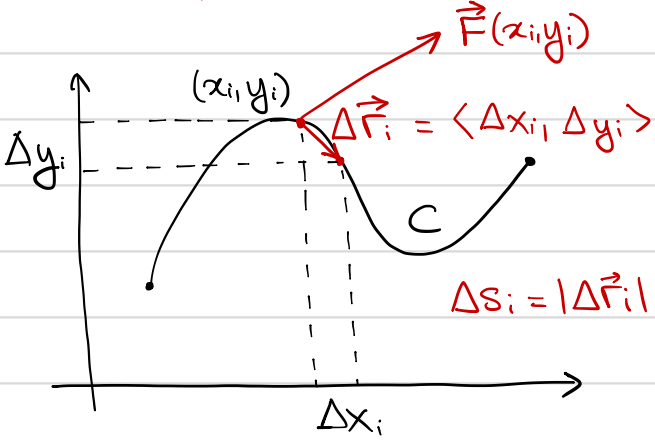
Application (2D):

$$\int_C f(x,y)\,ds = \text{Area of "fence"}$$



Line integrals of vector fields

$$\vec{F} = \langle P, \overset{R}{Q} \rangle$$



$$\vec{r}(t) = (x(t), y(t), \overset{z(t)}{\ })\ , \quad a \le t \le b$$

Definition (Work): $\vec{F} = \langle P, \overset{R}{Q} \rangle$

$$\underbrace{\int_C \vec{F}\cdot d\vec{r}}_{\text{"Work"}} := \lim_{\Delta r_i \to 0} \sum_i \vec{F}(x_i,y_i)\cdot \Delta\vec{r}_i$$

$$= \int_a^b F(\vec{r}(t))\cdot \frac{d\vec{r}}{dt}\,dt$$

Another way to write the same thing:

$$\int_C \overset{\langle P,Q\rangle}{\vec{F}} \cdot \overset{\langle dx,dy\rangle}{d\vec{r}} =$$

$$= \int_a^b \langle P, Q\rangle \cdot \overbrace{\left\langle \frac{dx}{dt}, \frac{dy}{dt} \right\rangle}^{\frac{d\vec{r}}{dt}}\,dt$$

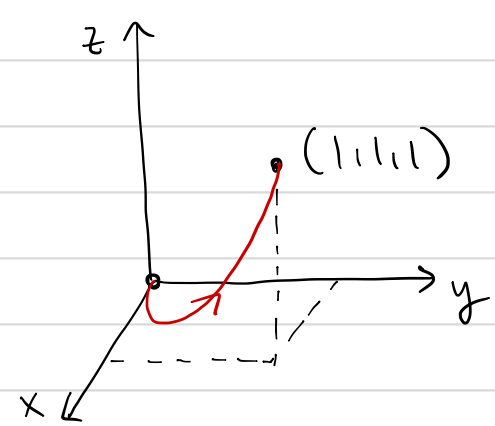
$$= \int_a^b \left(P\frac{dx}{dt} + Q\frac{dy}{dt} \right) dt$$

$$=: \int_C P\,dx + Q\,dy$$

Example:

C = twisted cubic curve

$$\begin{cases} x = t \quad y = t^2 \quad z = t^3 \\ 0 \le t \le 1 \end{cases}$$



$$\vec{F} = \langle \overset{P}{\underbrace{xy}}, \overset{Q}{\underbrace{yz}}, \overset{R}{\underbrace{zx}} \rangle$$

$$\int_C \vec{F} \cdot d\vec{r} =$$

$$= \int_C xy\,dx + yz\,dy + zx\,dz$$

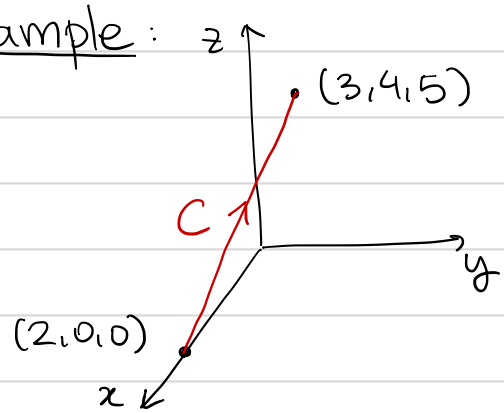
$$= \int_0^1 t^3 \cdot dt + t^5 \cdot 2t\,dt + t^4 \cdot 3t^2\,dt$$

$$= \int_0^1 t^3 + 5t^6 \; dt$$

$$= \frac{t^4}{4} + 5\frac{t^7}{7} \Big|_0^1$$

$$= \frac{1}{4} + \frac{5}{7} = \frac{7+20}{28} = \frac{27}{28} \;//$$

<u>Example</u>:



$$\vec{r}(t) = (2,0,0) + t\big((3,4,5) - (2,0,0)\big)$$
$$= (2+t,\ 4t,\ 5t)$$

$0 \leq t \leq 1$

$$\vec{F} = \langle y, z, x \rangle$$

$$\int_C \overset{\langle y,z,x\rangle}{\vec{F}} \cdot \overset{\langle dx,dy,dz\rangle}{d\vec{r}} =$$

$$= \int_0^1 \langle 4t, 5t, 2+t \rangle \cdot \langle dt, 4t, 5\,dt \rangle$$

$$= \cdots = \frac{49}{2} \;//$$

<u>Work = Change in kinnetic energy</u>

$\vec{r}(t)$ = position of particle at time t.

$\vec{F}(t)$ = net force on particle at time t.

$$= m \cdot \vec{a}$$
$$= m \cdot \vec{r}''$$

$$K = \frac{1}{2} m v^2 \qquad \text{(Kinetic energy)}$$

$$= \frac{1}{2} m\, \vec{r}' \cdot \vec{r}'$$

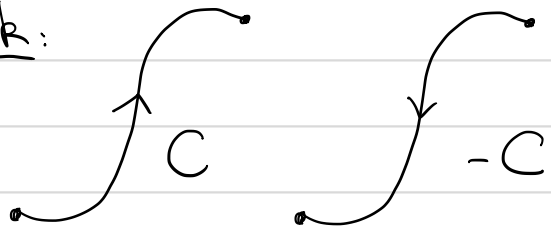
$$\frac{dK}{dt} = \frac{1}{2} m \left(\vec{r}'' \cdot \vec{r}' + \vec{r}' \cdot \vec{r}'' \right)$$

$$= m\, \vec{r}'' \cdot \vec{r}'$$
$$= \vec{F}(t) \cdot \vec{r}'(t)$$

Conclusion:

$$\underbrace{\int_a^b \vec{F}\cdot\frac{d\vec{r}}{dt}\,dt}_{W=\int_C \vec{F}\cdot d\vec{r}} = \underbrace{\int_a^b \frac{dK}{dt}\,dt}_{\Delta K}$$

<u>Remark</u>:



1) $\int_{-C} \vec{F}\cdot d\vec{r} = -\int_C \vec{F}\cdot d\vec{r}$

2) $\int_{-C} f\,ds = +\int_C f\,ds$